4.13.19

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Question

If the line $2x + y = k$ passes through the point which divides the line segment joining the points $1, 1$ and $(2, 4)$ in the ratio $3 : 2$, then k equals

1. $\frac{29}{5}$
2. 5
3. 6
4. $\frac{11}{5}$

Solution

Let the point **C** divide the line segment joining **A** and **B** in the ratio 3 : 2 where

$$\mathbf{A} = \begin{pmatrix} 1 \\ 1 \end{pmatrix} \tag{1}$$

$$\mathbf{B} = \begin{pmatrix} 2 \\ 4 \end{pmatrix} \tag{2}$$

$$\tag{3}$$

Formula

By section formula,

$$\mathbf{C} = \frac{\lambda \mathbf{B} + \mathbf{A}}{(\lambda + 1)}. \tag{4}$$

$$Here \left(\lambda = \frac{3}{2}\right) \tag{5}$$

The equation of the line can be represented as:

$$\mathbf{n}^{\mathbf{T}}\mathbf{x} = c_1 \tag{6}$$

$$where\ \mathbf{n}^{\mathbf{T}} = \begin{pmatrix} 2 & 1 \end{pmatrix}\ and\ c_1 = k. \tag{7}$$

Solution

As **C** lies on the above line,

$$\mathbf{n}^{\mathbf{T}}\mathbf{C} = c_1 \tag{8}$$

$$\begin{pmatrix} 2 & 1 \end{pmatrix} \frac{\lambda\mathbf{B} + \mathbf{A}}{(\lambda + 1)} = k \tag{9}$$

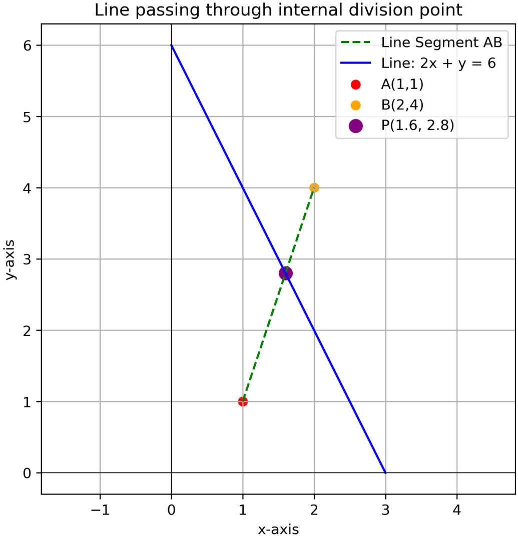
$$\frac{1}{5}\begin{pmatrix} 2 & 1 \end{pmatrix} (3\mathbf{B} + 2\mathbf{A}) = k \tag{10}$$

$$\frac{1}{5}\begin{pmatrix} 2 & 1 \end{pmatrix} \left(\begin{pmatrix} 6 \\ 12 \end{pmatrix} + \begin{pmatrix} 2 \\ 2 \end{pmatrix} \right) = k \tag{11}$$

$$\frac{1}{5}\begin{pmatrix} 2 & 1 \end{pmatrix} \begin{pmatrix} 8 \\ 14 \end{pmatrix} = k \tag{12}$$

$$k = \frac{16}{5} + \frac{14}{5} = 6. \tag{13}$$

Plot


Line passing through internal division point
Line Segment AB
Line: 2x + y = 6
A(1,1)
B(2,4)
P(1.6, 2.8)
y-axis
x-axis
6
5
4
3
2
1
0
−1
0
1
2
3
4

C Code

```c
#include <stdio.h>
// sol[0] = x, sol[1] = y, sol[2] = k
void find_k(double sol[3]) {

    double x1 = 1, y1 = 1;
    double x2 = 2, y2 = 4;
    int m = 3, n = 2;

    // Internal division point (x, y)
    double x = (m*x2 + n*x1) / (double)(m+n);
    double y = (m*y2 + n*y1) / (double)(m+n);

    double k = 2*x + y;

    // Store results
    sol[0] = x;
    sol[1] = y;
    sol[2] = k;
}
```

Python Code

```
import ctypes
import numpy as np
import matplotlib.pyplot as plt

# Call C library
lib = ctypes.CDLL(./findk.so)

# Define argument types
lib.find_k.argtypes = [ctypes.POINTER(ctypes.c_double)]

# Prepare result array
sol = (ctypes.c_double * 3)()
```

Python Code

```
# Call C function
lib.find_k(sol)

# Extract results
x, y, k = sol[0], sol[1], sol[2]

print(fPoint dividing AB in 3:2 = ({x}, {y}))
print(fValue of k = {k})

# Plot the graph
# Given points
A = np.array([1, 1])
B = np.array([2, 4])
P = np.array([x, y]) # Use point from C code
```

Python Code

```
# Line equation: 2x + y = k => y = k - 2x
x_vals = np.linspace(0, 3, 100)
y_vals = k - 2*x_vals

plt.figure(figsize=(6,6))

# Plot line segment AB
plt.plot([A[0], B[0]], [A[1], B[1]], 'g--', label='Line Segment
    AB')

# Plot line 2x+y=k
plt.plot(x_vals, y_vals, 'b', label=f'Line: 2x + y = {k:.0f}')

# Plot points
plt.scatter(*A, color='red', label='A(1,1)')
plt.scatter(*B, color='orange', label='B(2,4)')
plt.scatter(*P, color='purple', s=80, marker='o', label=f'P({x:.1
    f},{y:.1f})')
```

Python Code

```
# Labels
plt.xlabel('x-axis')
plt.ylabel('y-axis')
plt.title('Line passing through internal division point')
plt.axhline(0, color='black', linewidth=0.5)
plt.axvline(0, color='black', linewidth=0.5)
plt.legend()
plt.grid(True)
plt.axis(equal)

# Save the figure
plt.savefig(findk.png, dpi=300, bbox_inches='tight')
plt.show()
```